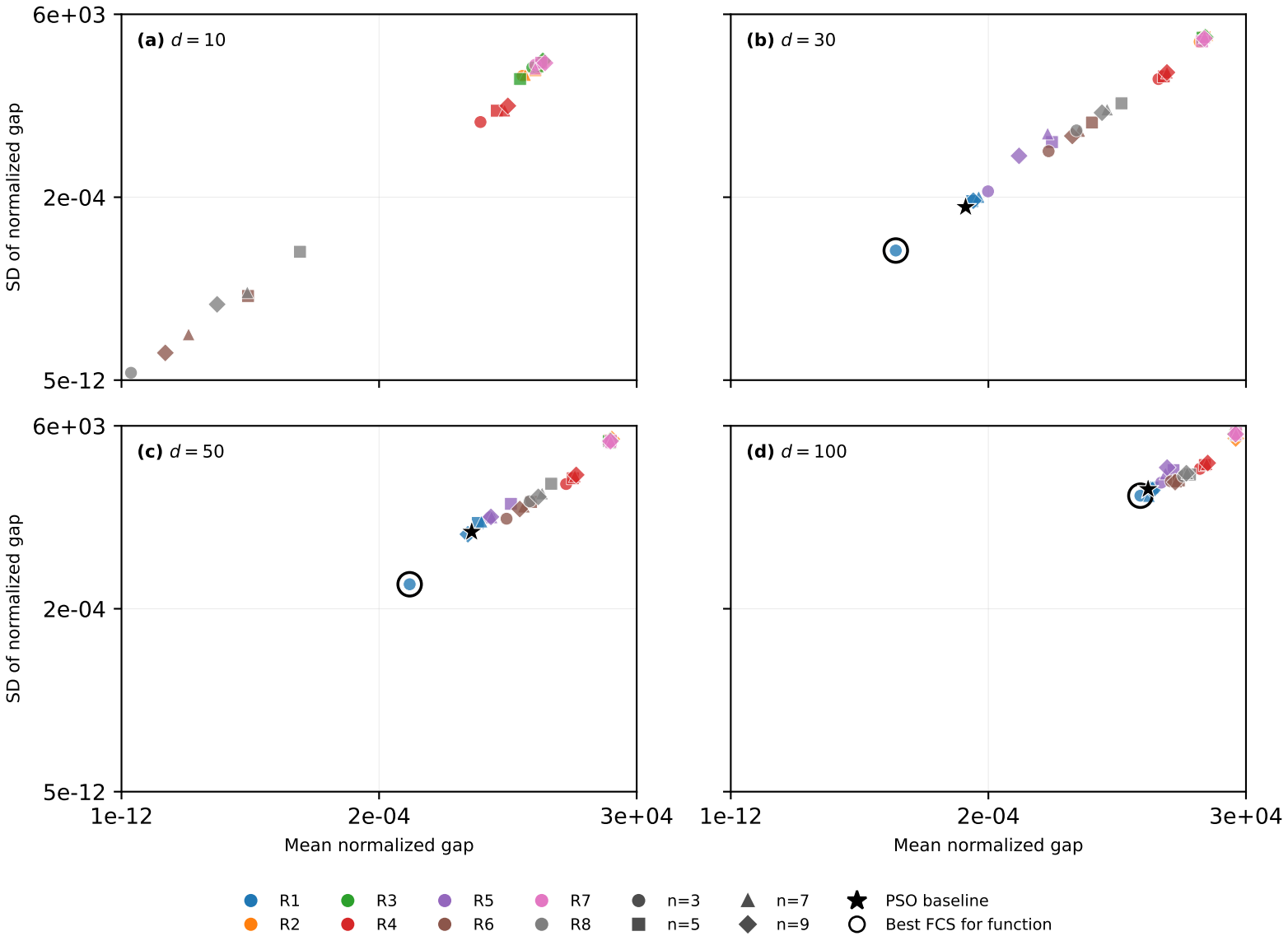
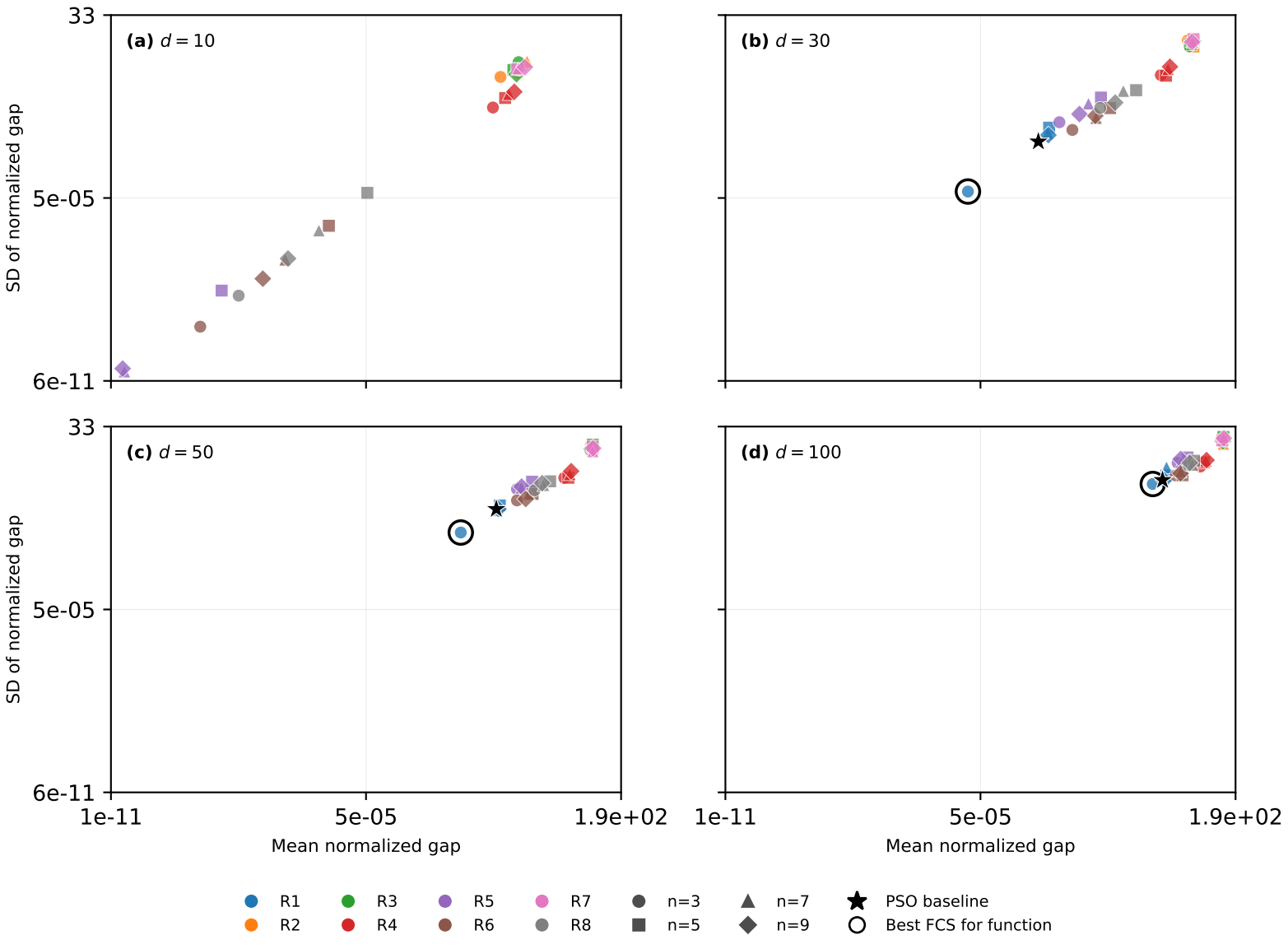
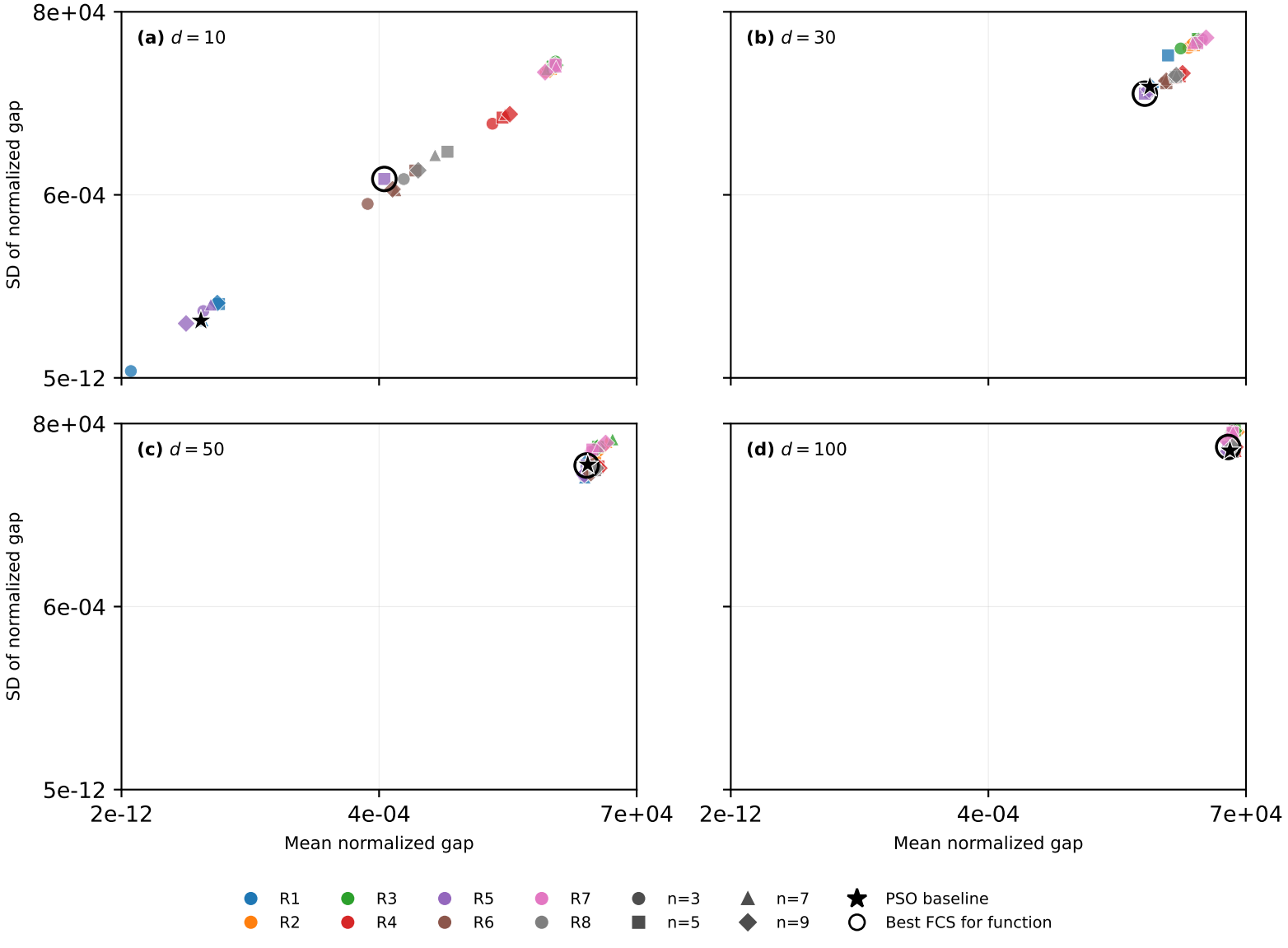
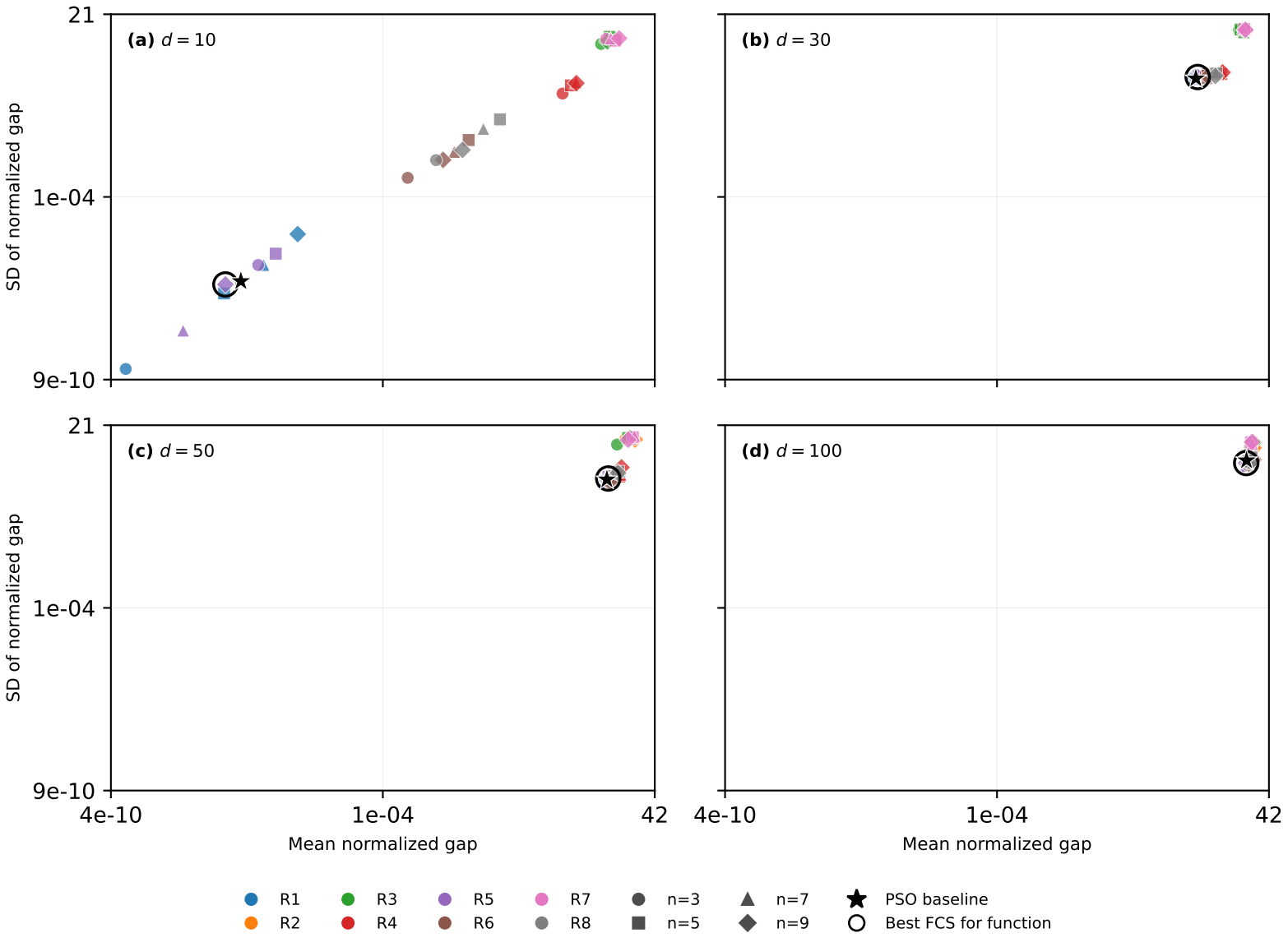
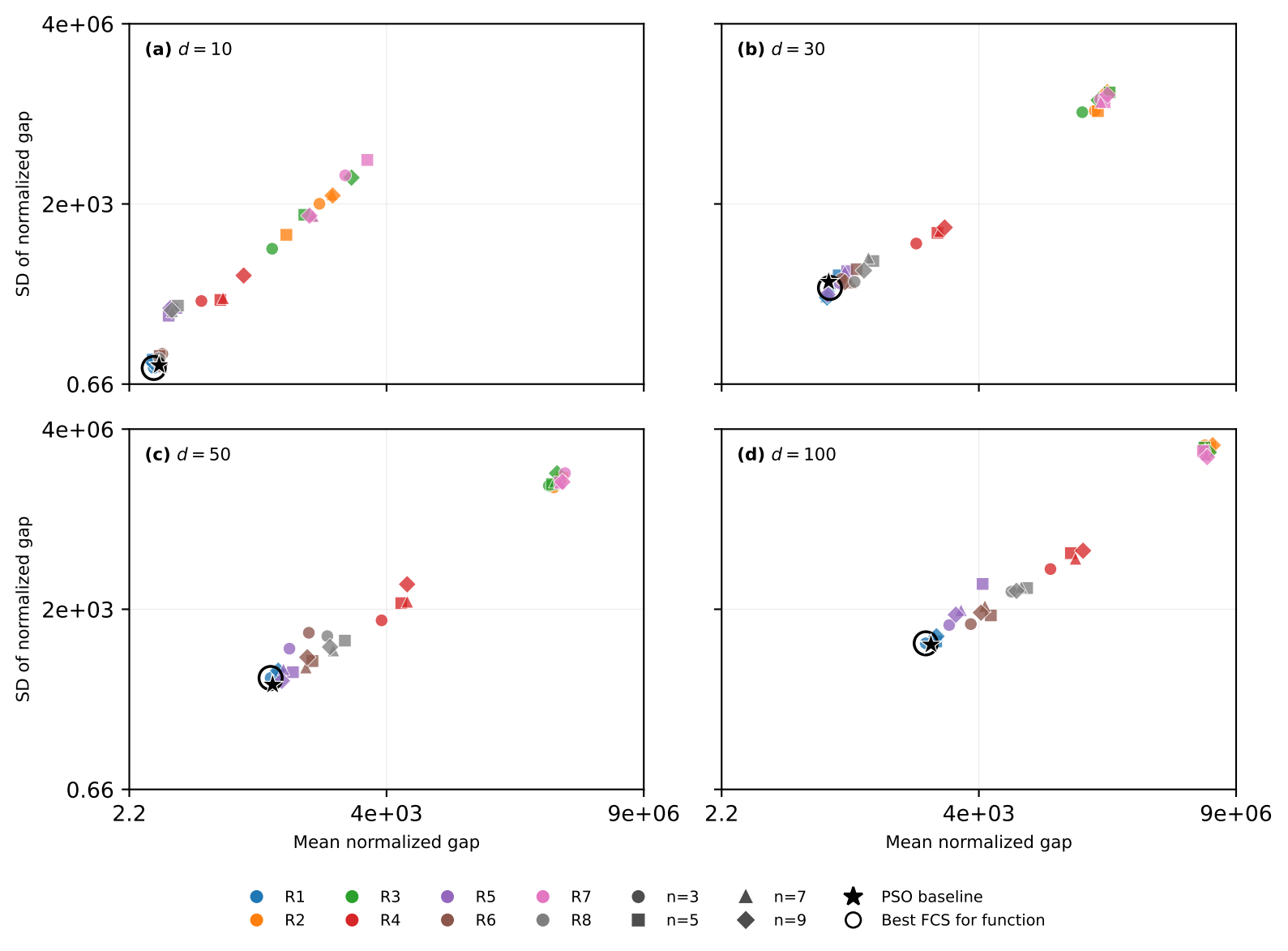
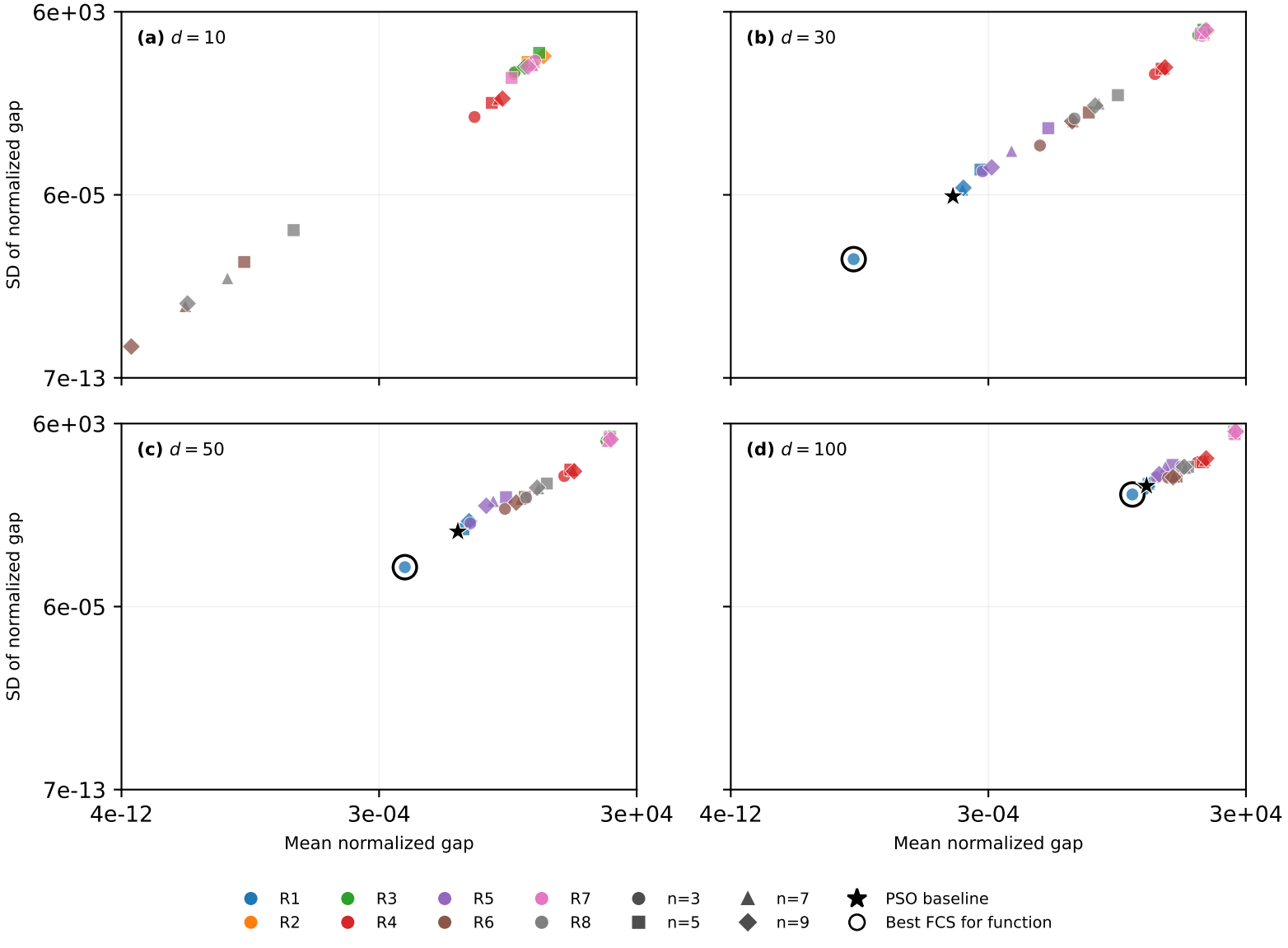

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
SD of normalized gap
Mean normalized gap
6e+03
2e-04
5e-12
1e-12
2e-04
3e+04
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

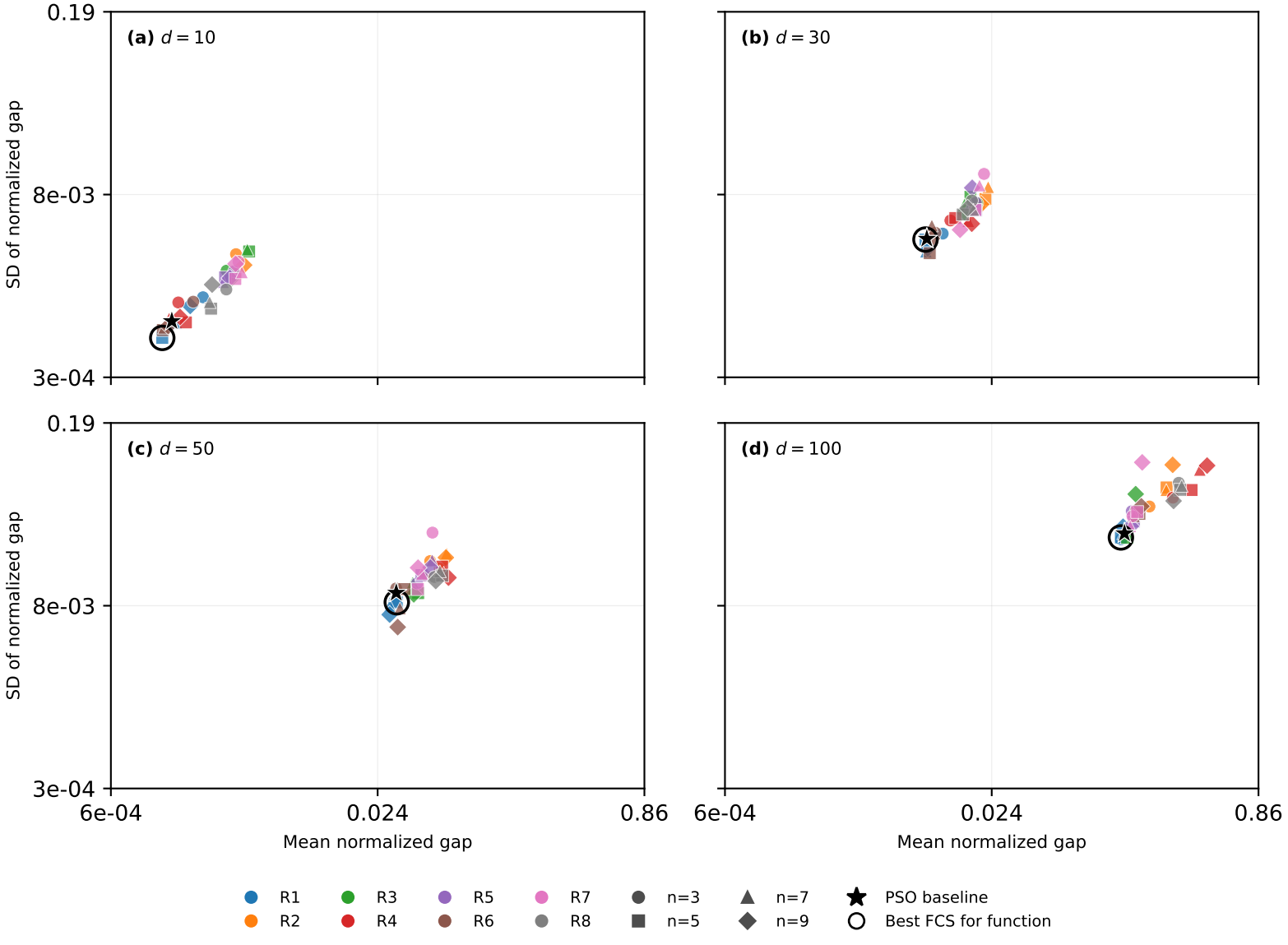

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
33
5e-05
6e-11
SD of normalized gap
1e-11
5e-05
1.9e+02
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

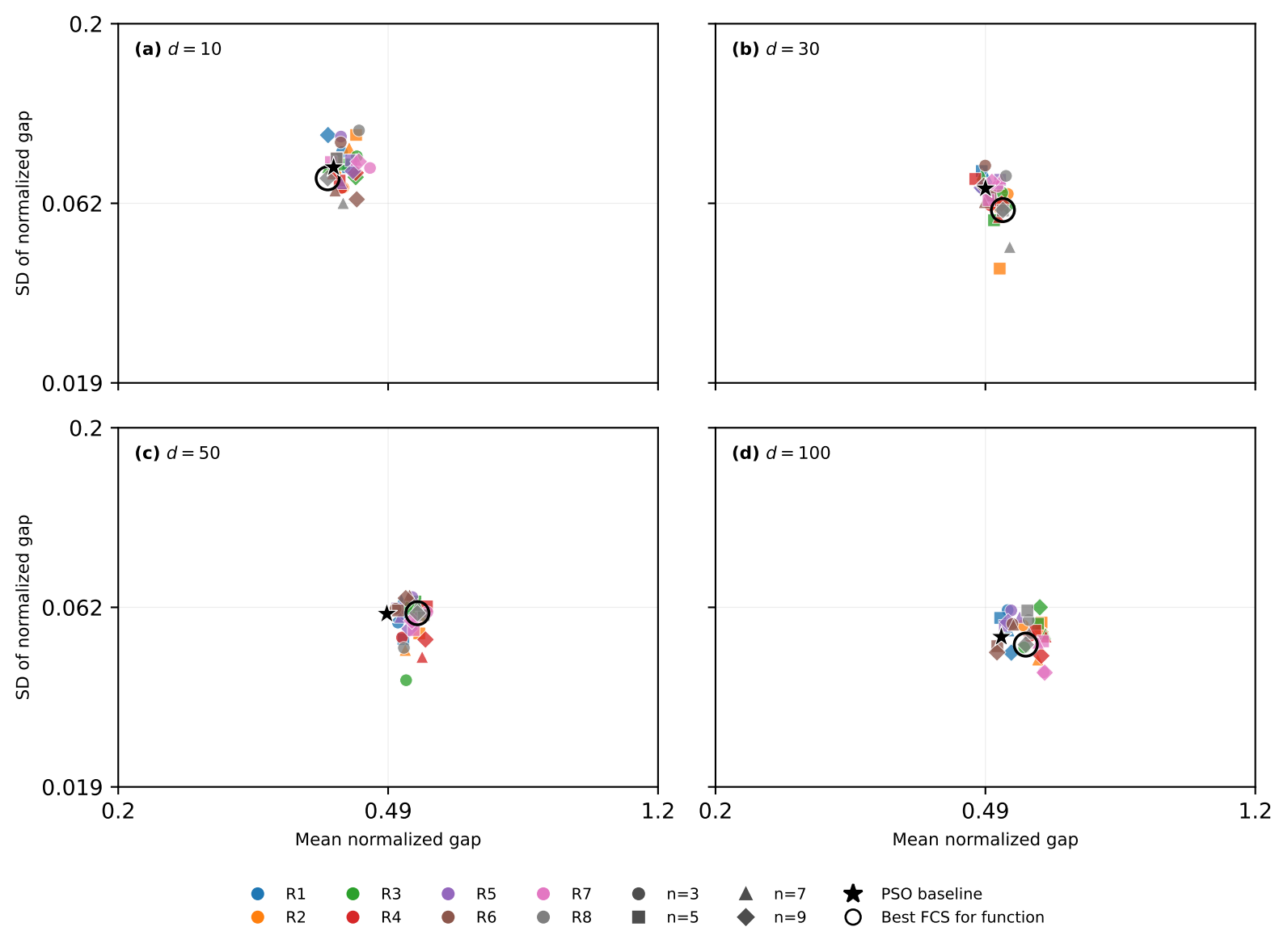

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
8e+04
6e-04
5e-12
SD of normalized gap
2e-12
4e-04
7e+04
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

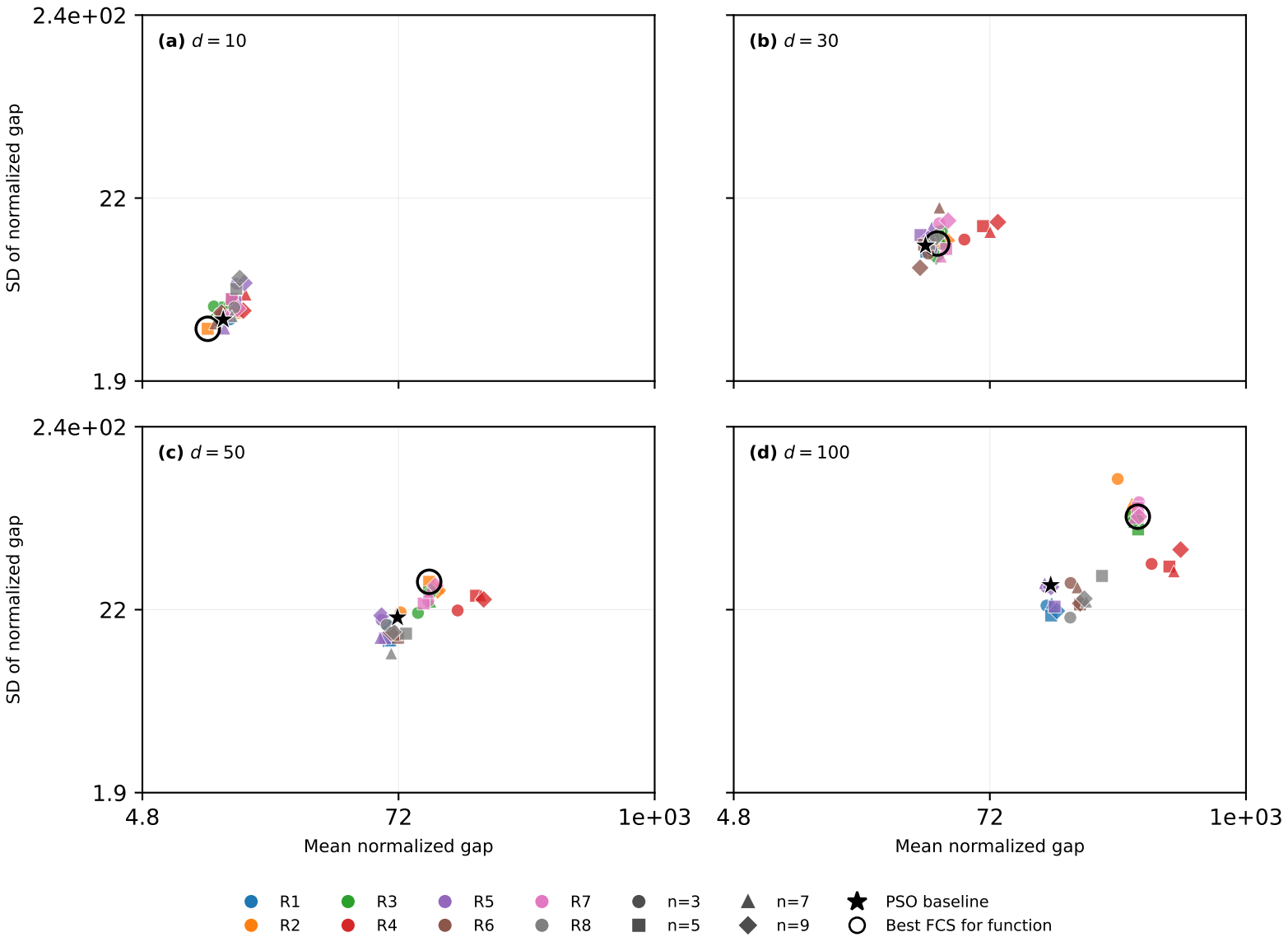

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
SD of normalized gap
21
1e-04
9e-10
4e-10
42
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

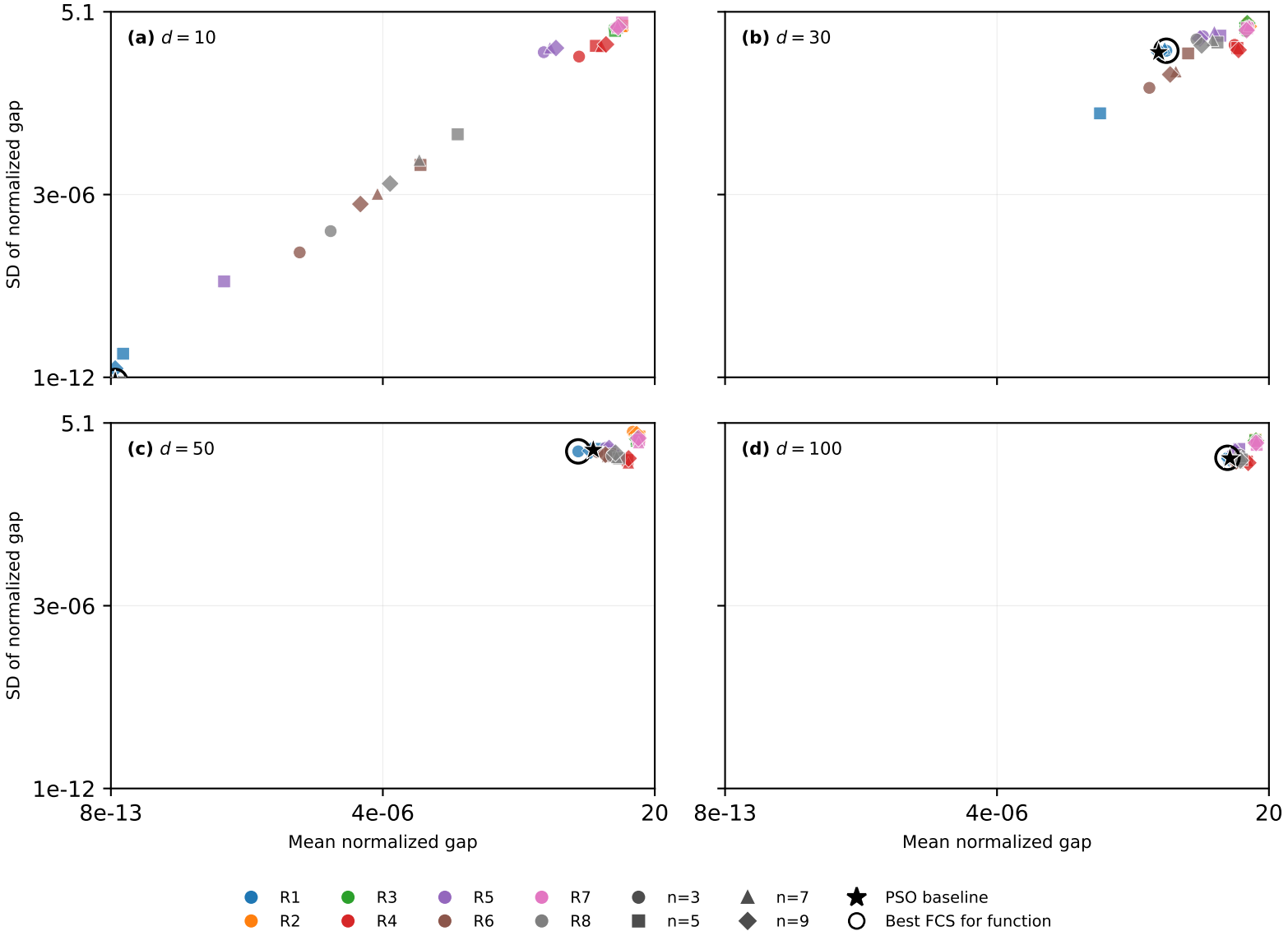

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
4e+06
2e+03
0.66
SD of normalized gap
2.2
4e+03
9e+06
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

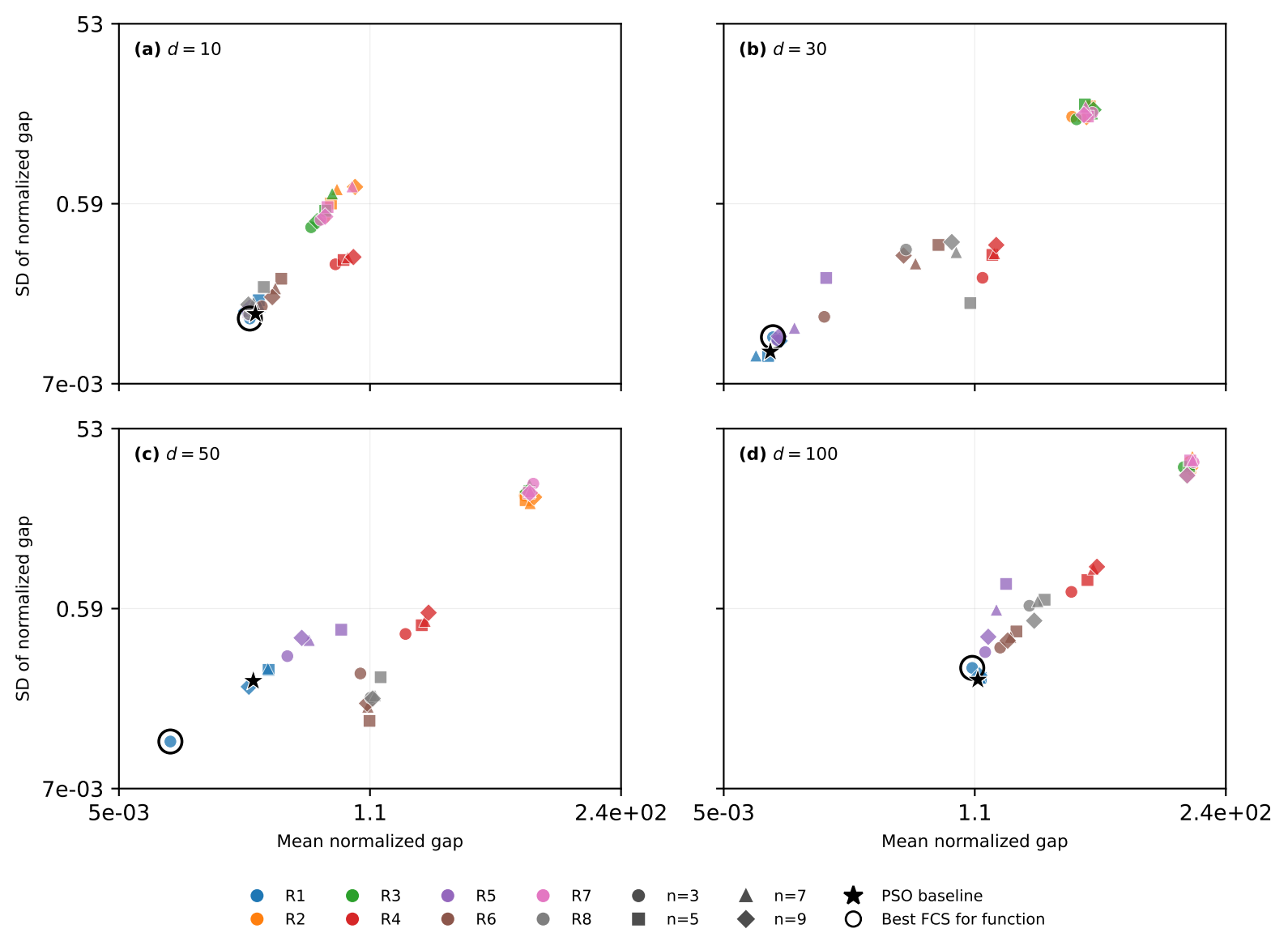

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
6e+03
6e-05
7e-13
SD of normalized gap
4e-12
3e-04
3e+04
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

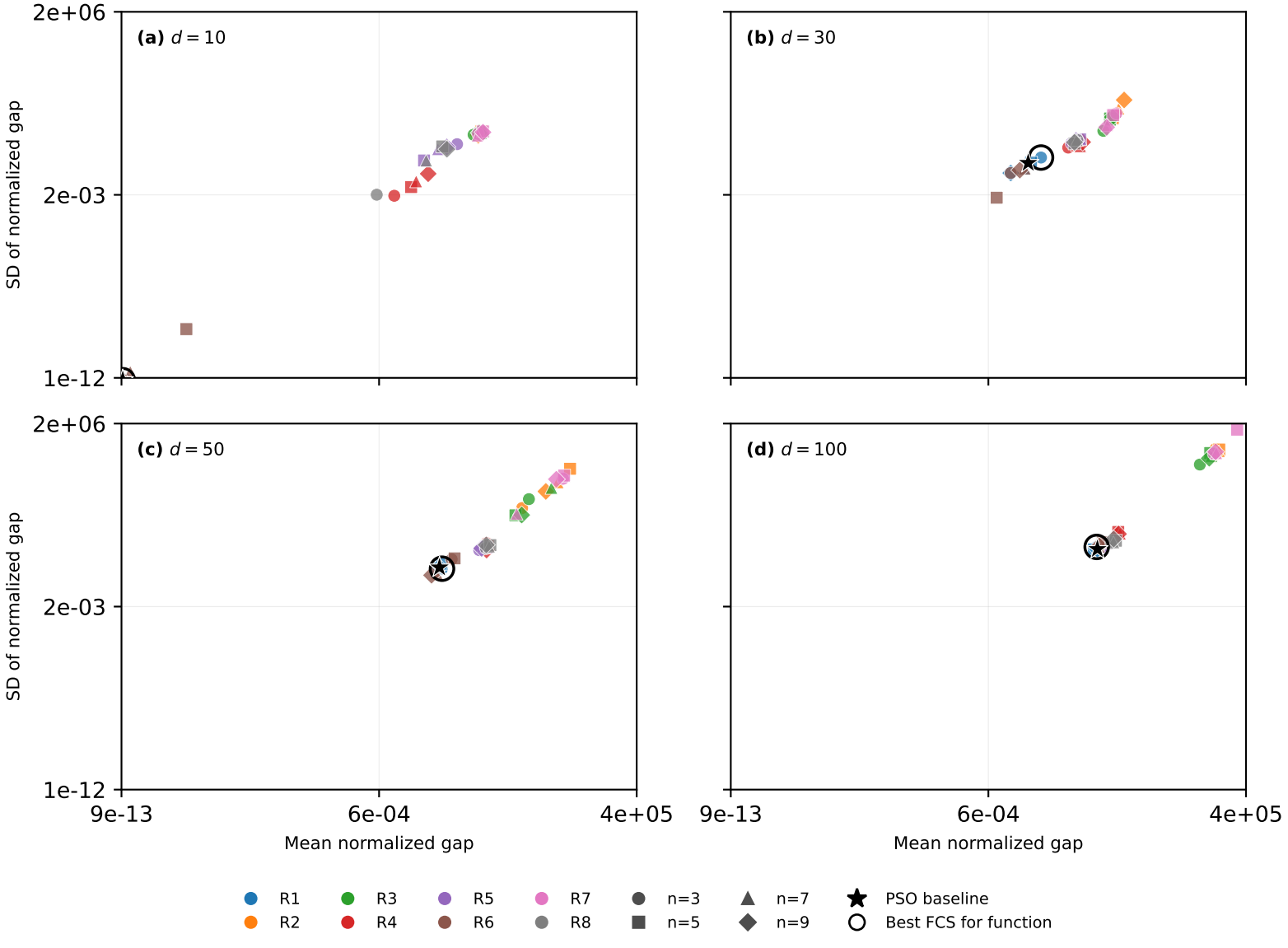

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
0.19
8e-03
3e-04
SD of normalized gap
6e-04
0.024
0.86
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

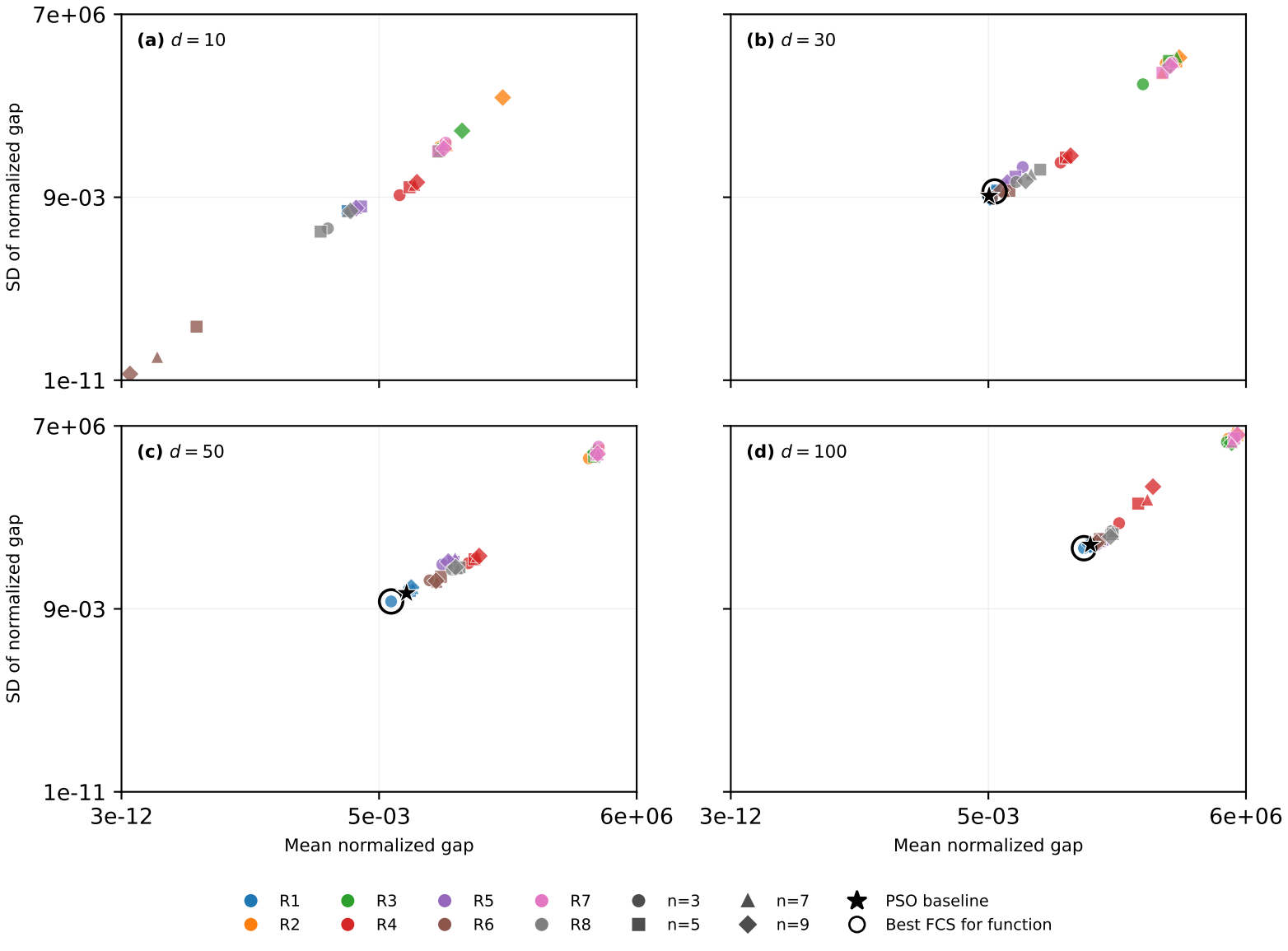

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
SD of normalized gap
Mean normalized gap
0.2
0.062
0.019
0.49
1.2
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

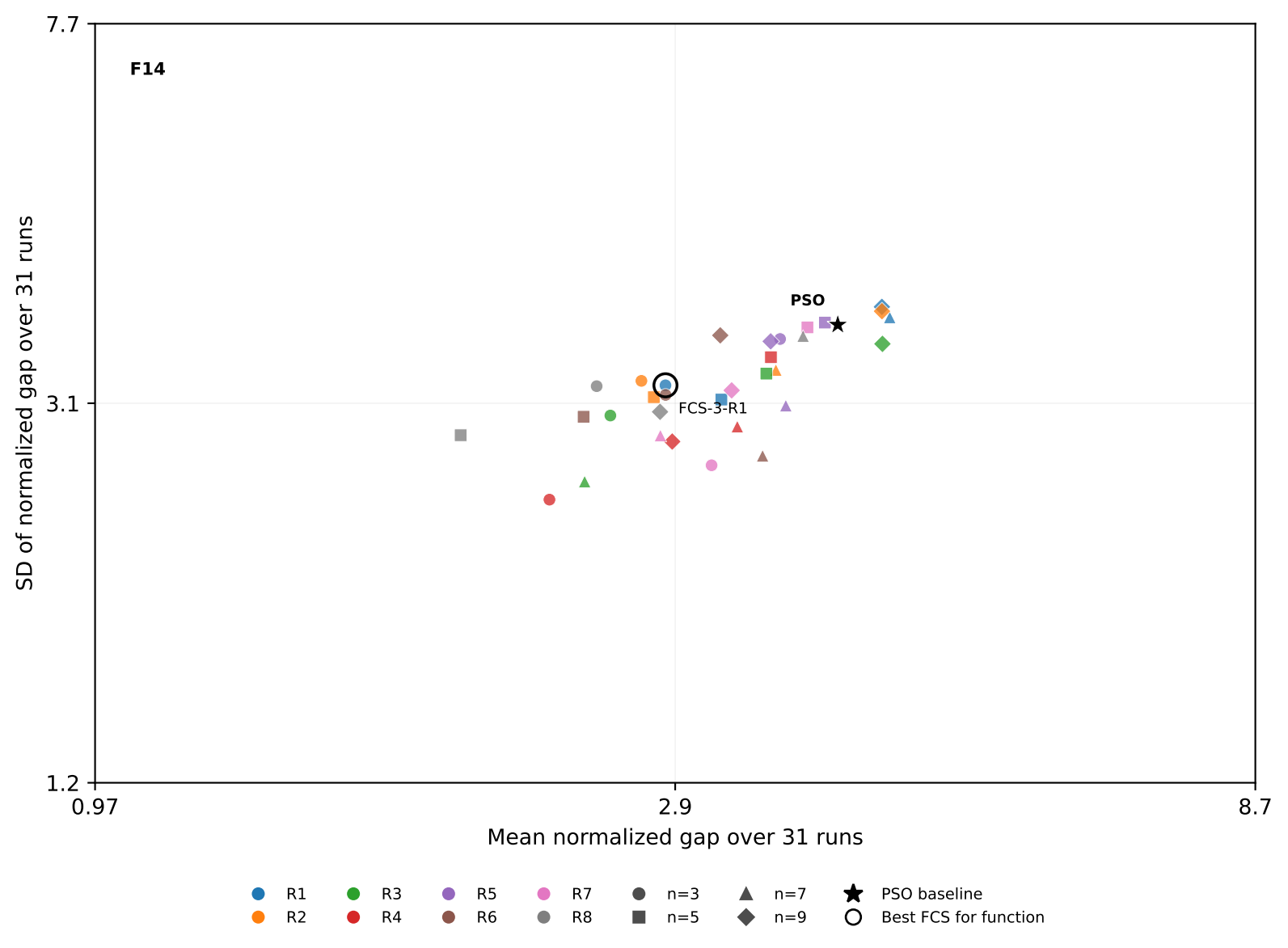

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
2.4e+02
22
1.9
SD of normalized gap
4.8
72
1e+03
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

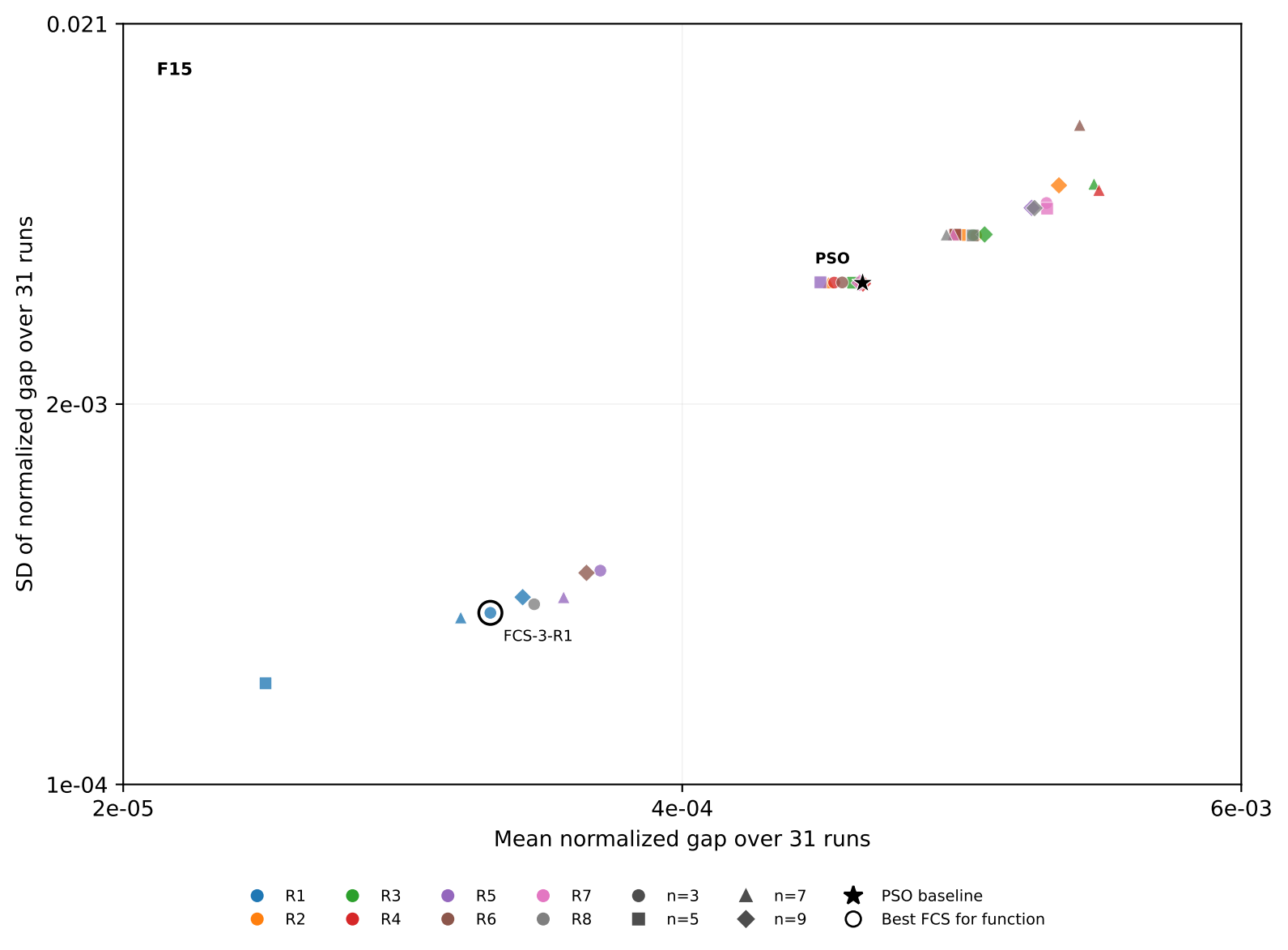

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
5.1
3e-06
1e-12
SD of normalized gap
8e-13
4e-06
20
Mean normalized gap
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

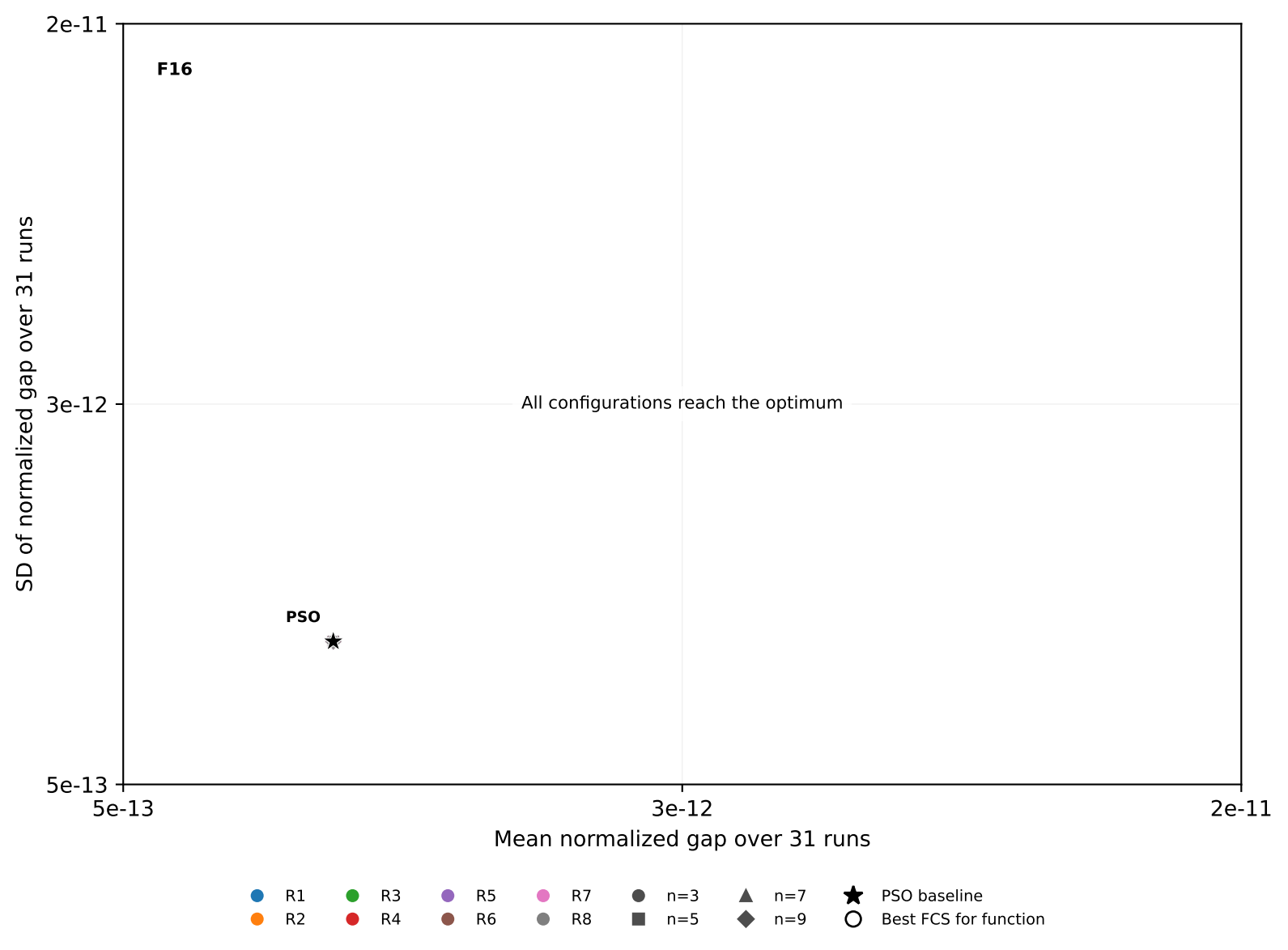

(a) $d = 10$
(b) $d = 30$
(c) $d = 50$
(d) $d = 100$
SD of normalized gap
Mean normalized gap
53
0.59
7e-03
5e-03
1.1
2.4e+02
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

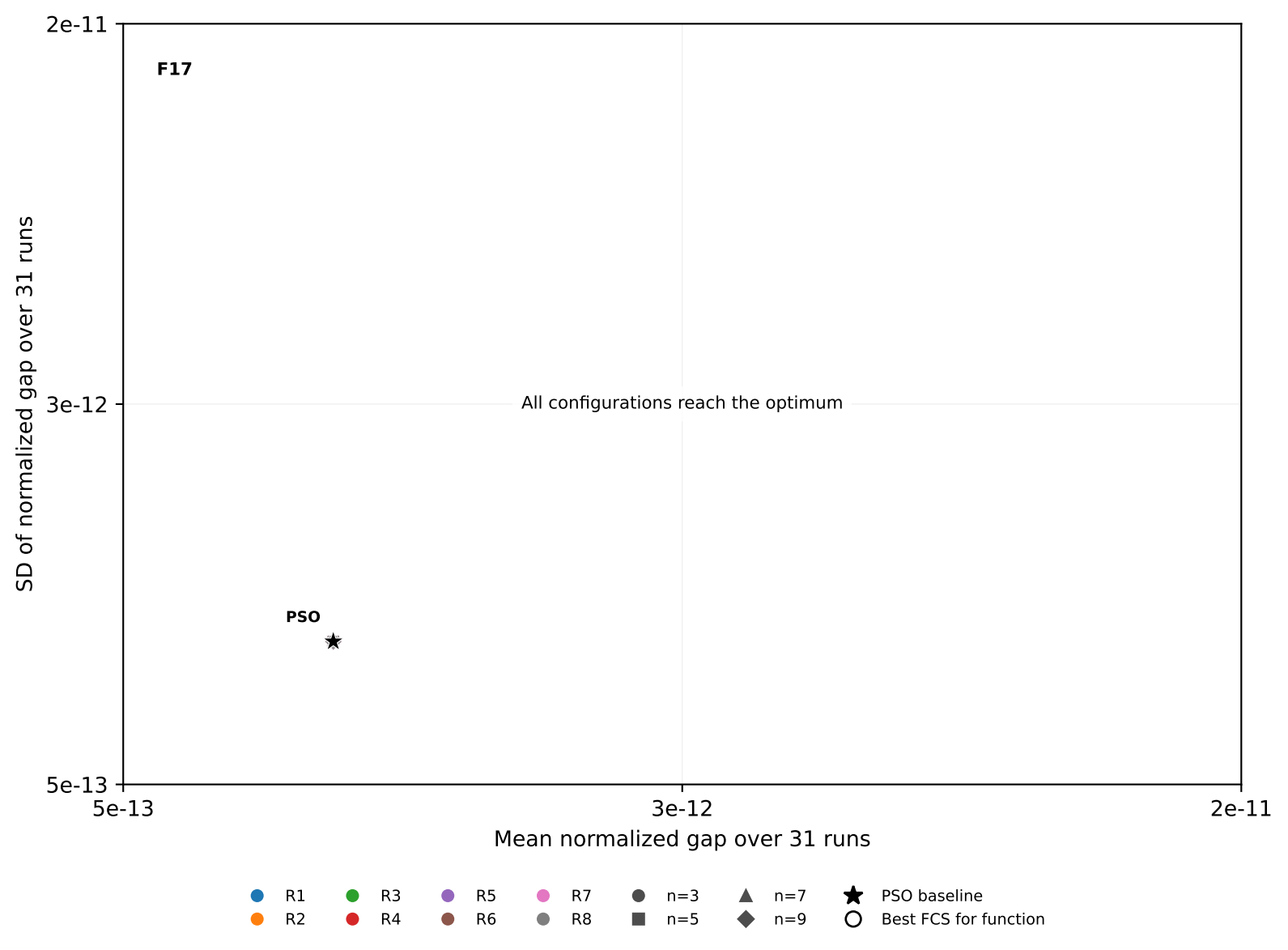

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
SD of normalized gap
2e+06
2e-03
1e-12
Mean normalized gap
9e-13
6e-04
4e+05
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

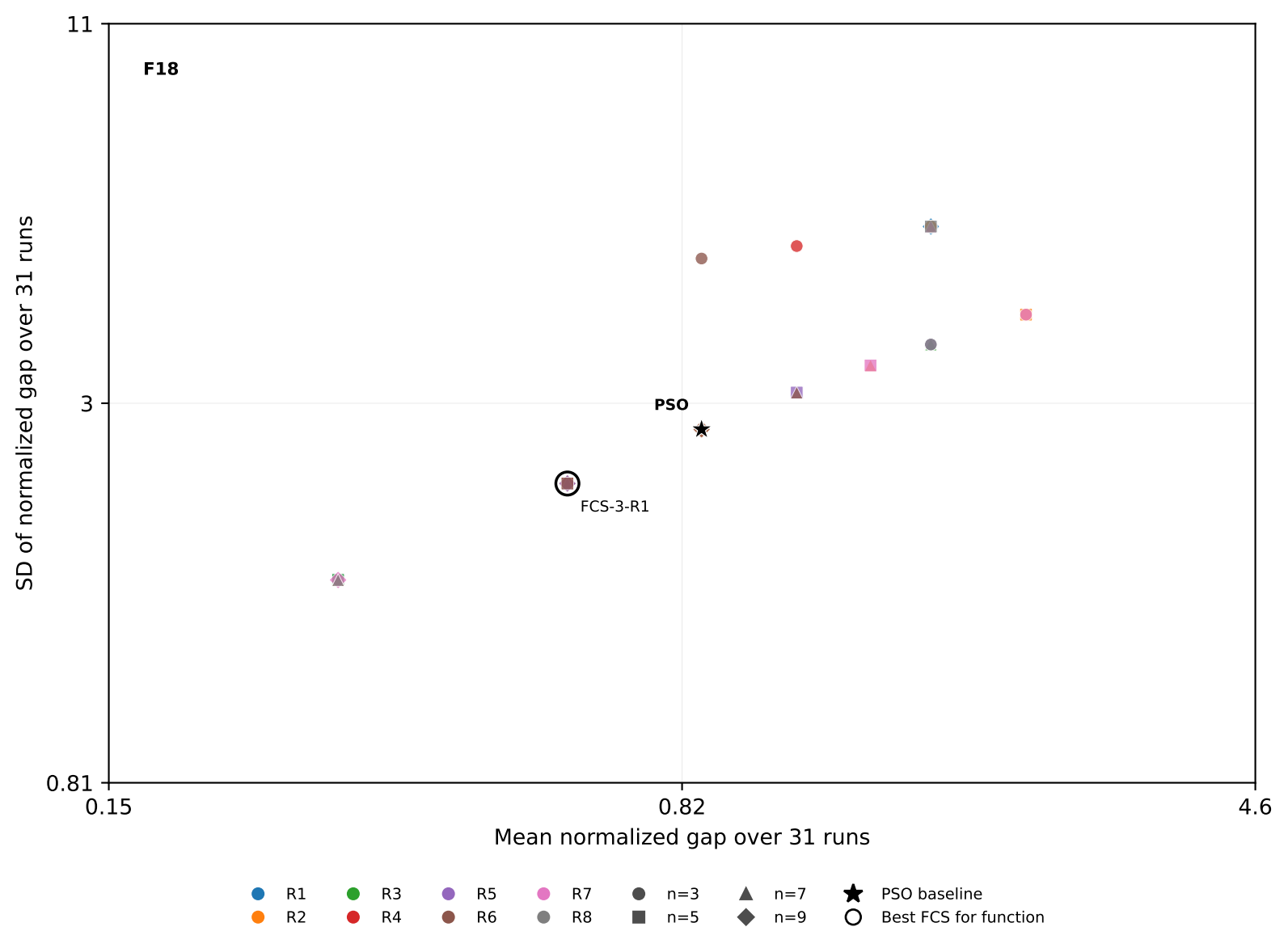

(a) d = 10
(b) d = 30
(c) d = 50
(d) d = 100
SD of normalized gap
Mean normalized gap
7e+06
9e-03
1e-11
3e-12
5e-03
6e+06
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

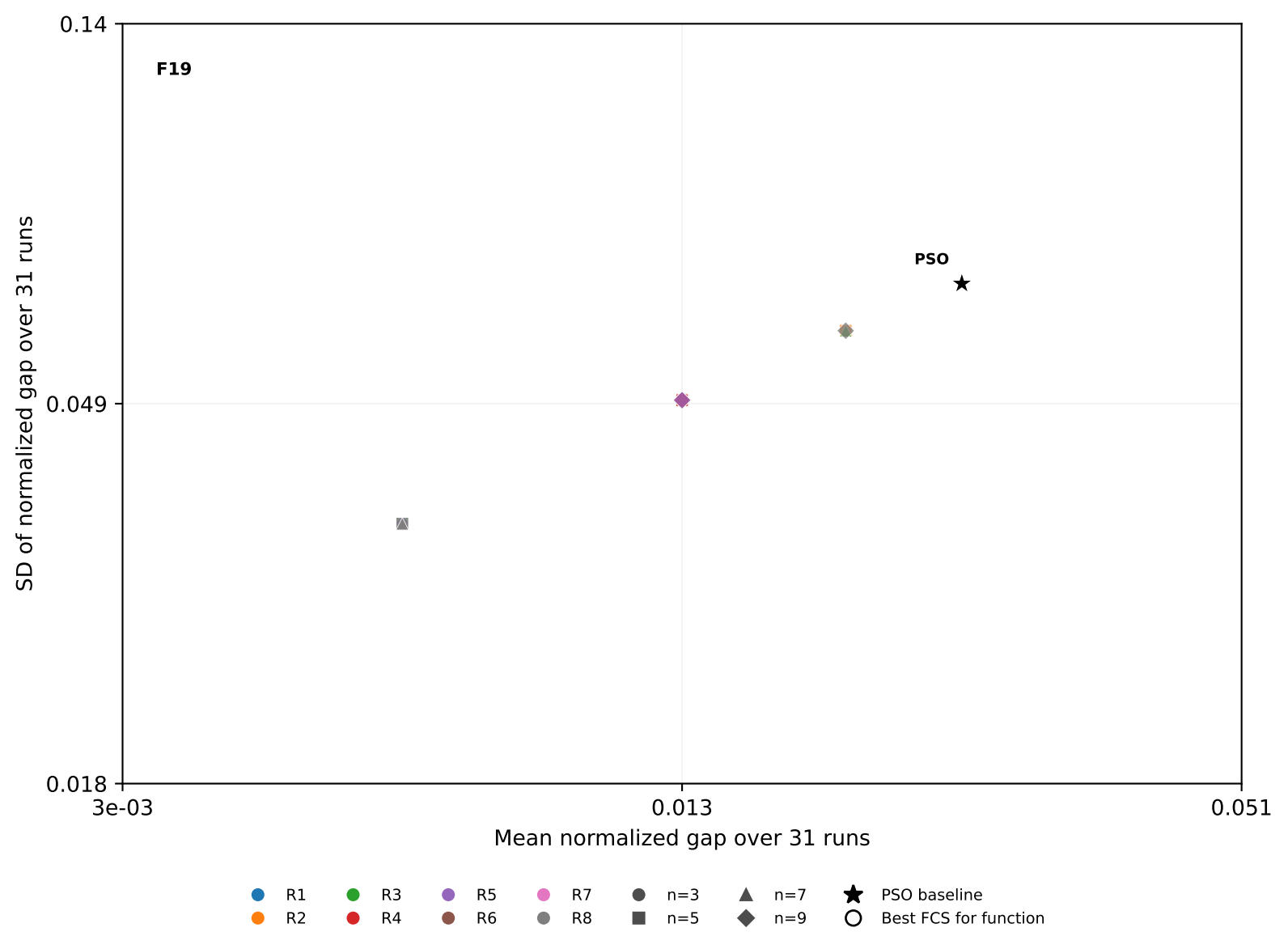

F14
PSO
FCS-3-R1
7.7
3.1
1.2
SD of normalized gap over 31 runs
0.97
2.9
8.7
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

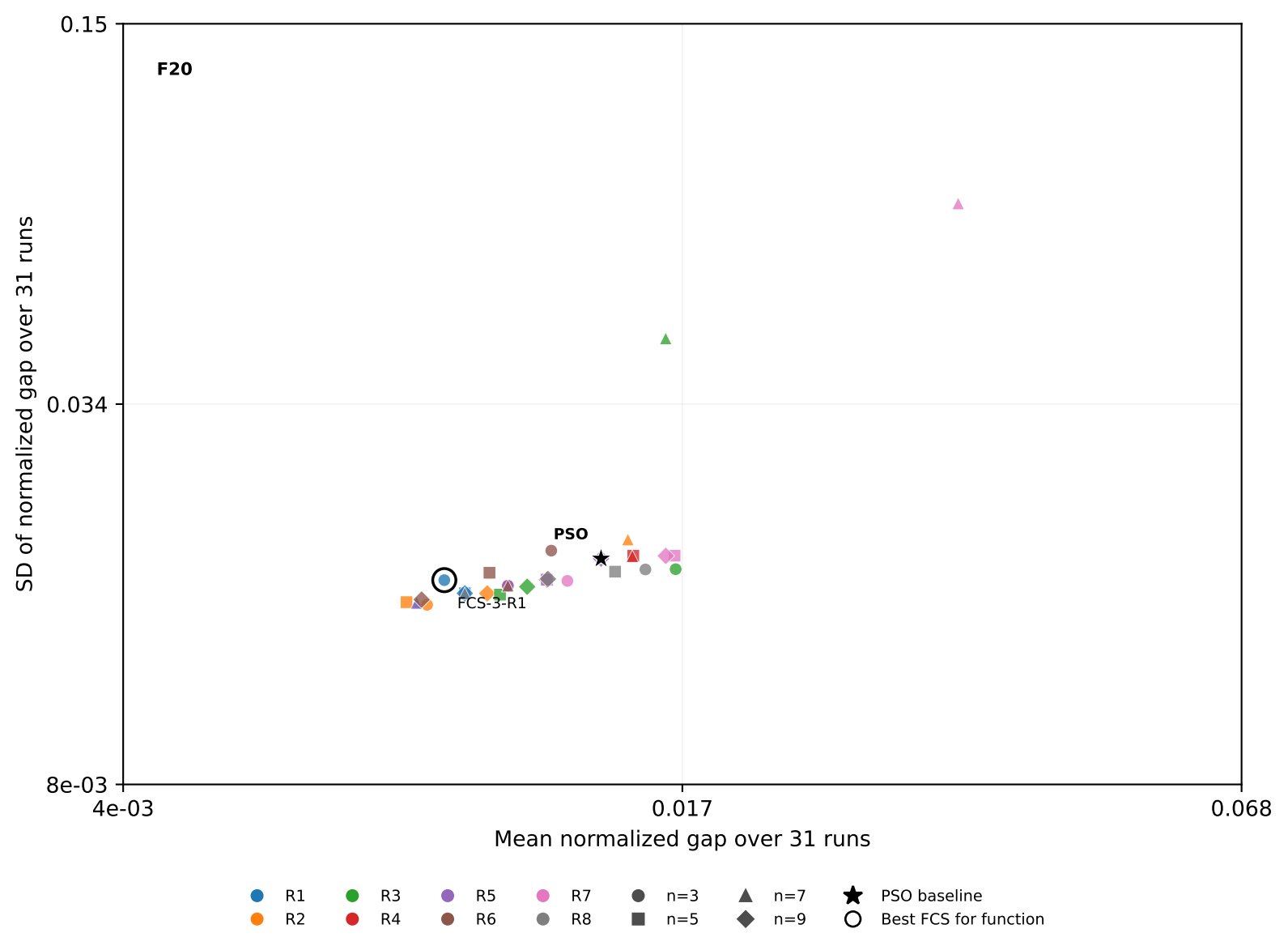

F15
PSO
FCS-3-R1
0.021
2e-03
1e-04
2e-05
4e-04
6e-03
SD of normalized gap over 31 runs
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

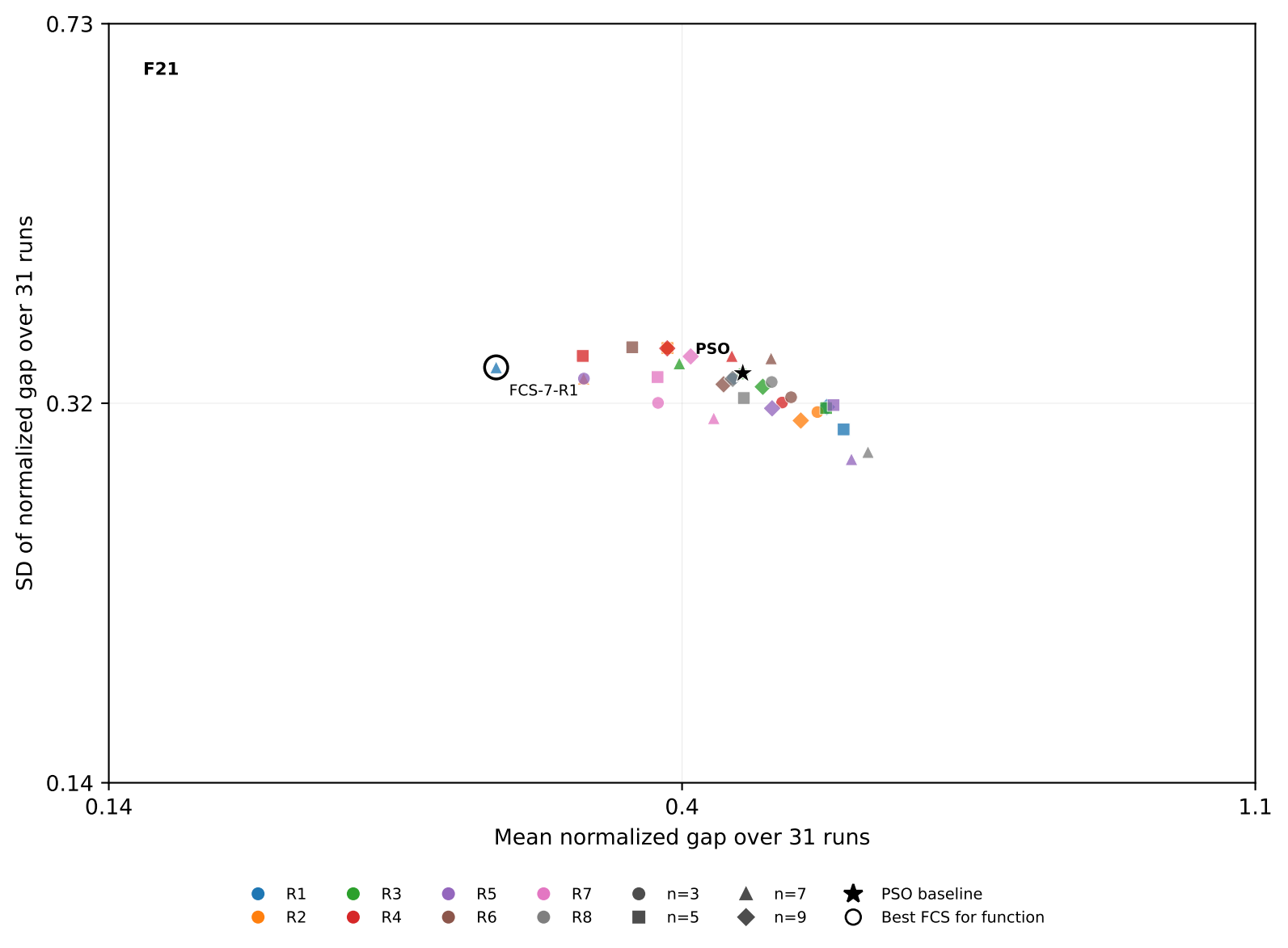

F16
All configurations reach the optimum
PSO
2e-11
3e-12
5e-13
SD of normalized gap over 31 runs
5e-13
3e-12
2e-11
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

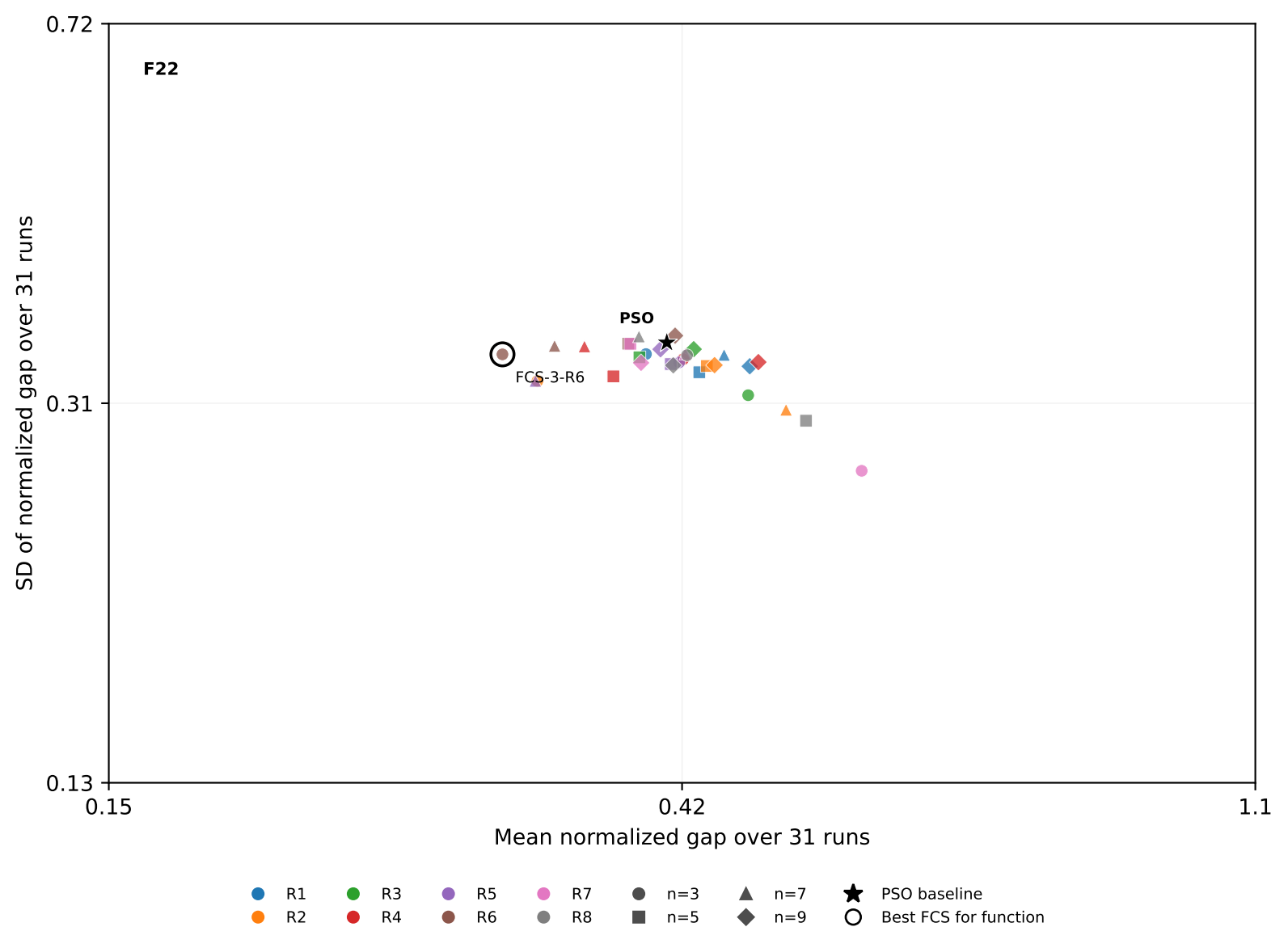

F17
All configurations reach the optimum
PSO
2e-11
3e-12
5e-13
SD of normalized gap over 31 runs
5e-13
3e-12
2e-11
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function

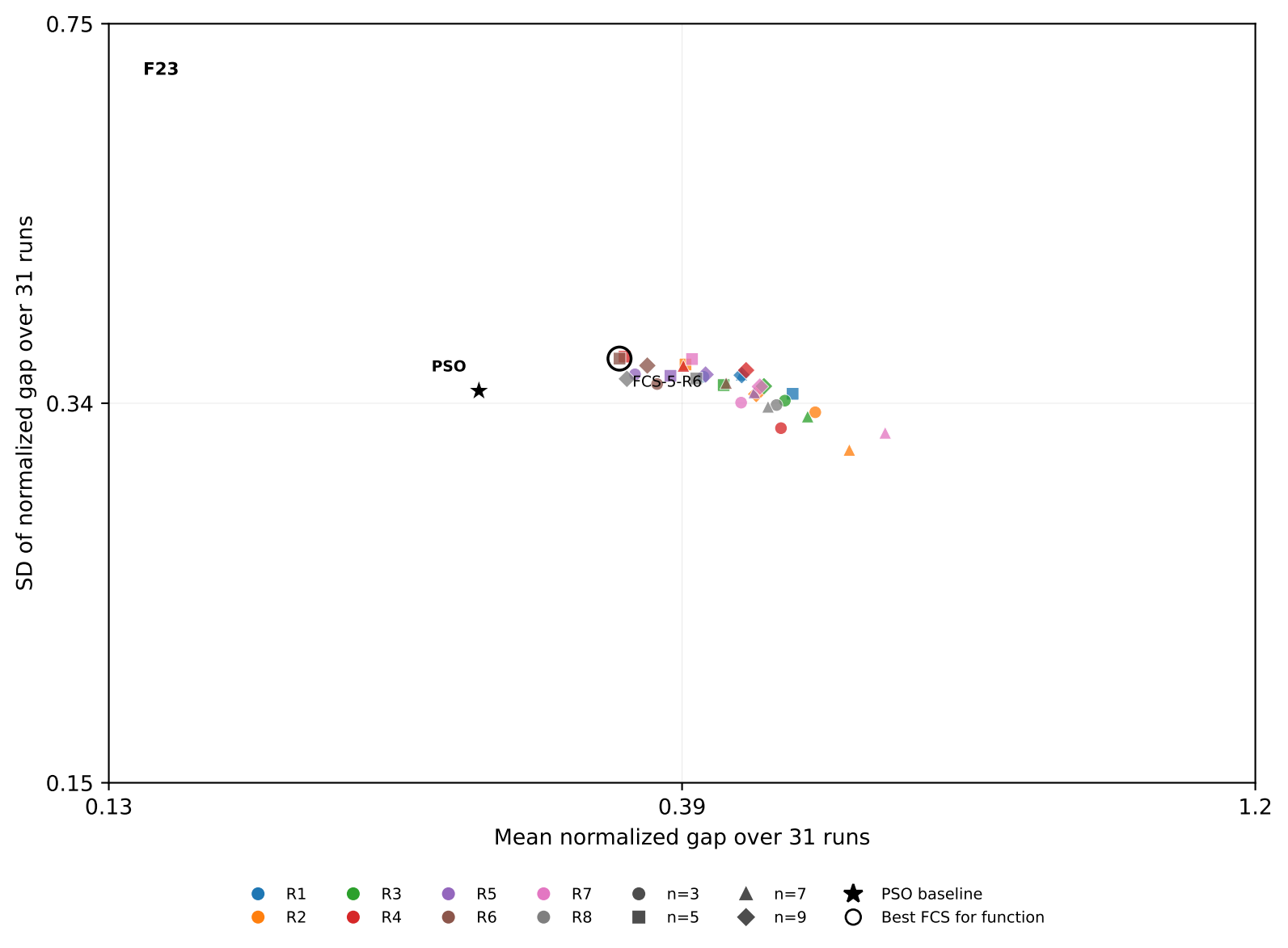

F18
11
3
0.81
SD of normalized gap over 31 runs
PSO
FCS-3-R1
0.15
0.82
4.6
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function


F19
PSO
0.14
0.049
0.018
3e-03
0.013
0.051
SD of normalized gap over 31 runs
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function


F20
0.15
0.034
8e-03
SD of normalized gap over 31 runs
PSO
FCS-3-R1
4e-03
0.017
0.068
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function


F21
PSO
FCS-7-R1
0.73
0.32
0.14
0.14
0.4
1.1
SD of normalized gap over 31 runs
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function


F22
PSO
FCS-3-R6
0.72
0.31
0.13
0.15
0.42
1.1
SD of normalized gap over 31 runs
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function


F23
PSO
FCS-5-R6
0.75
0.34
0.15
0.13
0.39
1.2
SD of normalized gap over 31 runs
Mean normalized gap over 31 runs
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function